You said:

Python/Django Hiring Assignment: AI-Assisted Box Selection System
Context
We operate an ecommerce platform. When a customer places an order, the warehouse team needs to know which shipping box should be used. Each product has dimensions and weight. Each box has internal dimensions, maximum weight capacity, and cost.
Your task is to design and build a small Django-based system that recommends the most suitable box for an order.
You are allowed to use AI tools such as ChatGPT, Claude, Cursor, GitHub Copilot, or similar tools. However, your submission must clearly show your own thinking.

ChatGPT said:

You said:

Context
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Your task is to design and build a small Django-based system that recommends the most suitable box for an order.
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Act as a senior Python developer reviewing a hiring assignment. Read the attached PRD carefully. Summarize the business problem, functional requirements, expected inputs and outputs, technical constraints, deliverables, and evaluation criteria. Separate mandatory requirements from optional improvements. Identify ambiguities and

ask questions before proposing an implementation. Do not write code yet.

Worked for 13s >

ChatGPT said:

You said:

Based on the PRD, recommend a minimal, maintainable solution using Python, Django, Django REST Framework, and SQLite if appropriate. Explain the responsibilities of each component, the API contract, database model, validation strategy, and selection algorithm. Avoid adding a frontend or unnecessary dependencies unless required by the assignment.

ChatGPT said:

You said:

Design a Django model for shipping boxes with a unique name, length, width, height, maximum weight, and cost. Recommend suitable field types, decimal precision, constraints, and migrations. Explain the trade-offs and provide code only after explaining the design.

ChatGPT said:

You said:

Implement a pure Python service that determines whether a product fits inside a shipping box. Consider all orientations of the three dimensions and enforce the maximum weight. Then select the cheapest suitable box. Define deterministic tie-breaking behavior and explain its assumptions. Include edge cases and unit tests.

ChatGPT said:

You said:

Design a Django REST Framework serializer for product length, width, height, and weight. Reject missing, malformed, zero, and negative values, and handle decimal inputs consistently. Explain appropriate error responses and avoid accepting invalid or ambiguous measurements.

ChatGPT said:

You said:

Implement a POST endpoint at /api/recommend/ using Django REST Framework. Validate the request, retrieve available shipping boxes, call the selection service, and return a clear JSON response for success, invalid input, and no suitable box. Keep business logic outside the view and explain the request-response flow.

ChatGPT said:

You said:

Create Django tests for cheapest-box selection, dimension rotation, overweight products, dimensions that fit no box, invalid inputs, empty box inventory, and equal-cost tie-breaking. Explain each test's purpose and expected result. Use the project's actual API and model names.

Worked for 26s >

ChatGPT said:

You said:

Review my existing implementation and actual test output. Identify concrete bugs, warnings, incorrect assumptions, and missing edge cases. Distinguish confirmed defects from optional improvements. Suggest the smallest safe changes, and never claim a test passes unless its output confirms it.

Worked for 40s >

ChatGPT said:

You said:

Help me document this implementation with a README, AI usage disclosure, test cases, and test output. Use simple, clear language. Derive technical details from the actual code and test results. Clearly mark anything that still needs verification, and do not invent test results or past AI prompts give manualy

Analyzed

ChatGPT said:

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README.md
Document

AI_USAGE.md
Document

TEST_CASES.md
Document

TEST_OUTPUT.md
Document

You said:

Review my completed repository against the original hiring assignment. Check the code, documentation, setup instructions, tests, GitHub Actions or accepted test-output evidence, AI usage disclosure, and all required deliverables. Produce a checklist of completed items, missing items, and risks. Do not mark anything complete without evidence.

Worked for 25s >

ChatGPT said: